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A novel approach to modeling and visualisation of epidemic outbreaks: combining manual and automatic calibration

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***Abstract*—In this paper, we present an approach which couples a calibration algorithm for a compartmental model of infection dynamics with a graphical user interface which allows to visualise the modeling results and to simulate control measures aimed at curtailing the spread of infection. Unlike the most popular approaches related to visualisation of disease dynamics, our method simultaneously allows to obtain the parameter values based on epidemic data and change these values in interactive mode. The simulation tool built according to this principle is intended to help the interdisciplinary teams to obtain parameter values based on objective and subjective (domain-specific) criteria. We demonstrate the usage of the resulting modeling and visualisation framework using influenza outbreak data in Saint Petersburg for two epidemic seasons.**

***Index Terms*—epidemiology, influenza, Python, optimization**

I. Introduction

Decision taking during an unfolding epidemic scenario is a complex and diverse task that requires the most accurate and reliable mathematical and statistical methods [1]. These methods imply the collection and visualisation of epidemic data, as well as applying models calibrated to that data and interpreting simulation results, made by collective efforts of IT and healthcare specialists. The results of such analysis are often ambiguous due to uncertainty and errors which accompany data collection and selection of models [2]. Beside that, the criteria for the optimal model parameters and for the most effective control measures based merely on quantifiable measurements might be limited from the point of epidemiologists and policymakers, thus the control effort with high theoretical effectiveness might not correspond to the best decision in a particular real-life situation. Thus, it might be convenient to allow the domain specialists to manipulate the model parameters to tweak the optimal set of values obtained by automatic calibration of the model. In this paper, we present an approach which couples a calibration algorithm for a differential model of infection dynamics with a graphical user interface which allows to visualise the modeling results and to simulate the effects of various interventions, such as social distancing, quarantine measures, or vaccination campaigns. Unlike the most popular approaches related to visualisation of disease dynamics, our method simultaneously allows to obtain the parameter values based on epidemic data and change these values in interactive mode. According to our idea, the approach proposed in this study will help specialists from different domains to collaborate more efficiently and, as a result, to facilitate a multidisciplinary challenge of curtailing the infection spread.

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II. Related works

Graphical user interfaces dedicated to epidemic modeling and decision support are developed and described by numerous research teams. J. Dykes et al. [3], mentioned the merits of an interactive interface in epidemic modeling, and developed recommendations. Norazaliza Mohd Jamil et al. described the virtues of GUI-based epidemic modeling to better understand the situation by non-domain specialists [4] . In addition, there are also works that consider the creation of GUI applications for visualizing several models simultaneously [5], [6] to perform more detailed outbreak analysis. Judith V. Douglas et al. [7] presented a modeling-visualisation framework that allows to model control measures. For the purposes described above, software frameworks and methodologies can be used for more effective development of application dashboards in education, science or business [8], [9].

III. Methods

A. Data

The dataset used in the study was based on the data provided by the Research Institute of Influenza. By combining ARI incidence data for Saint Petersburg, Russia, and the results of virological testing, we found the numbers for weekly clinically registered cases of ARI attributed to different influenza strains (A(H1N1)pdm09, A(H3N2), B) [2] and age groups (0–2, 3–6, 7–14, and 15+) [10]. For the sake of data uncertainty reduction, age groups from 0 up to 14 years old were aggregated and used in the study as a common group. Provided data includes cases of infection in 2010-2020 along with markers for weeks which correspond to the period of officially declared influenza epidemic. Although in our previous study [11] only one season was used in the experiments, we assume that the described algorithm is versatile enough to be applied to any other epidemic season in a given Russian city which has sufficient volume of reported incidence and virological testing. Based on the findings from our prior scientific work [10], which involved an assessment of the data quality during each epidemiological season, two specific seasons, namely 2010-2011 and 2015-2016, were chosen for further analysis.

B. Model and automatic calibration

In modeling framework we use a SEIR–type discrete compartmental model represented by a system of difference equations, which allows to distinguish between various age groups and strains. The model is described in detail in our previous study [11]. We use automatic calibration which implies application of two optimization algorithms, namely simulated annealing and L-BFGS-B.

The model calibration error is calculated as the distance between the incidence data ($Z^{(\mathrm{dat})}$) and the simulated incidence values ($Z^{(\mathrm{mod})}$):

$$F\left(Z^{(\mathrm{mod})}, Z^{(\mathrm{dat})}\right) = \sum_{i=0}^{t_1} w_i \cdot \left(z_i^{(\mathrm{mod})} - z_i^{(\mathrm{dat})}\right)^2 \longrightarrow \min . \tag{1}$$

The weights w_i included in the formula (1) are used to reflect the 'importance' of fitting the model curve to the particular data points. In case of seasonal influenza, the data closer to the peak have higher plausibility, because the laboratory sample sizes tend to become bigger during a full-fledged outbreak and thus the proportion of different virus strains in the overall ARI incidence is calculated more precisely. Thus, for weight estimation we use the formula $w_i = \sigma^{-d}$, where d is the distance of the current data point from the peak in time units (weeks), σ is a parameter related to decreasing rate with growing distance from the peak.

Such quality metric as a coefficient of determination R^2 was estimated to evaluate a goodness of the model fit. Nonetheless, it is possible that this metric might not adequately capture the quality of the calibrated model, leading to the requirement of subjective expertise for a more precise representation of disease dynamics. In the subsequent chapter, we introduce our tool, which aims to address this limitation by integrating specialized knowledge from experts. This approach seeks to rectify the disparity associated with the metric and offer a more accurate depiction of the underlying disease dynamics. Furthermore, we present a modification to the model structure that permits the asynchronous start of different circulating strains within the population.

C. GUI and manual calibration

The results derived from the optimization procedure described previously serve as the initial parameter values for the following manual fine-tuning process. By employing this two-step approach, we aim to effectively address the model's sensitivity to parameter fluctuations and attain more reliable and precise parameter estimates that align with real-world epidemiological dynamics.

1) Motivation: Epidemic dynamic is usually governed by numerous intertwined factors, resulting in a complex and high-dimensional parameter space that encompasses transmission rates, latent and infectious periods, contact intensities, background immunity levels, and more. This complex landscape poses challenges for optimization algorithms, as they must explore a vast search space to identify the best-fitting parameters. Furthermore, uncertainties and noise presented in the real epidemic data badly influences the convergence of the automatic calibration algorithm. Data obtained during the early stages of an outbreak might be sparse or unreliable, that makes it challenging for automated optimization algorithms to accurately capture the dynamics of the epidemic, resulting in suboptimal model fits. Real epidemic data is often aggregated from diverse sources, each with its inherent limitations and biases.

Human expertise and intuition play a crucial role in narrowing down feasible parameter ranges, interpreting contextual nuances, and incorporating domain-specific knowledge. Our graphical user interface (GUI) is designed to facilitate the manual interactive calibration, which aims to enhance the model performance with incorporated expert knowledge. Such combination of the capabilities of optimization algorithms with the insights human experts incorporated into the proposed framework allows to improve reliability and relevance of epidemic modelling for informed decision-making. By adjusting the infectivity rates, which govern the likelihood of transmission between susceptible and infectious individuals, researchers can simulate the effects of various interventions. Similarly, modifying the proportion of immune individuals, which reflects either the process of natural infection or the influence of vaccination, allows for the exploration of herd immunity thresholds and the potential effectiveness of different immunization strategies. Hence, specific parameters like the proportion of immune individuals and infectivity rates are integrated into the GUI, allowing for manual refinement of the model. This facilitates the monitoring of the dynamic relationship between control measures and the trajectory of the epidemic. The GUI allows users to carefully adjust the parameters while observing immediate changes in the model's behavior, providing a valuable opportunity to assess the impact of small parameter perturbations.

2) Employment: Unlike in optimization process, the part of the framework adopted for the GUI addresses the forward problem [12]. The forward problem involves the specification of initial conditions and a predefined set of parameters, followed by the computation of the model's response. In this context, the GUI serves as a tool to simulate the model's behavior based on the provided parameter values and ascertain the corresponding outcomes.

- **Software.** Regarding the software applied for the GUI development, we used Python as a programming language. Dash [13], [14], [15], a Python framework, was employed for the construction of the graphical interface.

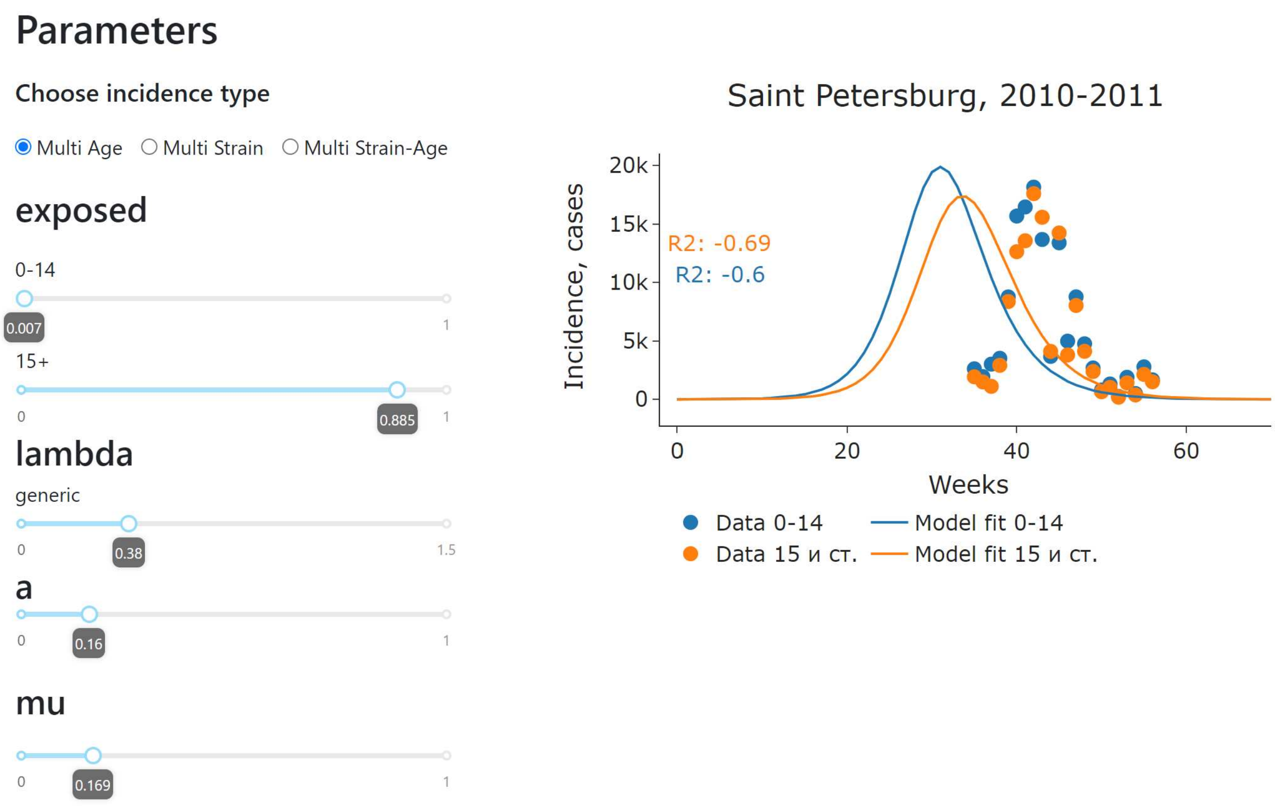


Fig. 1. Graphical user interface for manual model calibration

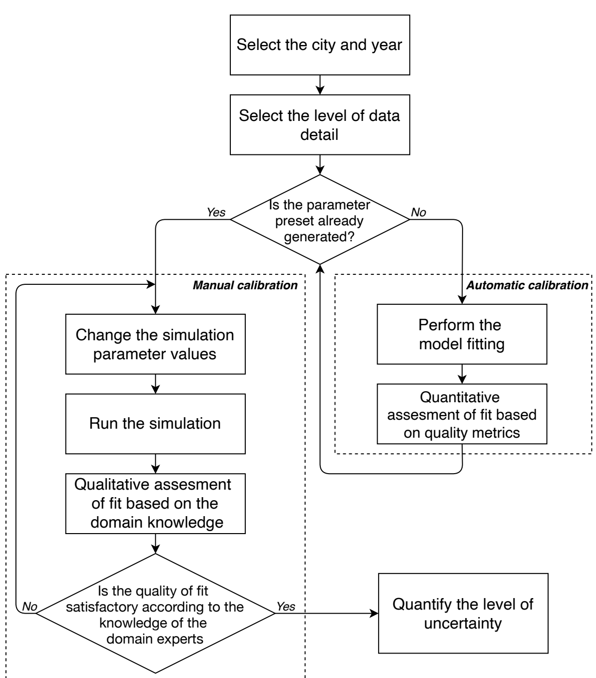


Fig. 2. A scheme of the modeling-visualisation framework

The screenshot demonstrating the current version of the interface is presented in Figure 1.

- **Inputs and outputs.** The input parameters employed for user interface corresponded to those listed in Table 1. Nevertheless, the inclusion of more detailed data leads to a fluctuation in the number of unidentified parameters. To address this, an abstraction of callbacks was exploited, facilitating the tracking and utilization of values derived from dynamically changing interface components. If standard components allowing numerical value input were utilized as components for parameter values, the output component represented a graph for visualisation of the model fit and original data. The scheme of the resulting multicomponent framework is demonstrated in Figure 2.
- **Future vision.** The present iteration of the interface facilitates manual calibration, allowing users to choose the level of data granularity, thereby affording them the opportunity to conduct comprehensive analysis pertaining to the model's data fit and quality. In addition to manual calibration, several features will be incorporated into the GUI to enhance its functionality for epidemiological model analysis. One such feature is the capability to generate short- and long-term model forecasts. Users can input different intervention scenarios and control measures, and the GUI will provide real-time predictions of the epidemic's trajectory over various time horizons

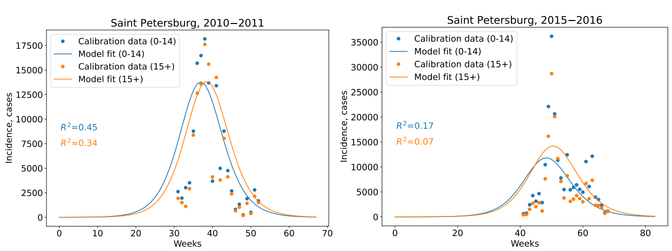
along with uncertainty quantification.

TABLE I
GUI INPUT PARAMETERS FOR MODEL FITTING

Definition	Description
α	Individuals with the history of exposure in the previous season
β	Intensity of infectious contacts
a	Fraction of previously exposed individuals who will lose immunity in the following epidemic season
μ	Fraction of population which do not participate in the infection transmission
δ	Shift between original incidence data and simulated curve

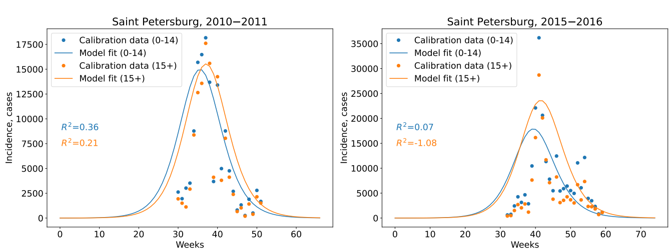
IV. Use cases

In this section, we present the results of experiments with the model calibration process over incidence data of two epidemic seasons in 2010-2011 and 2015-2016. Figure 3 illustrates the model's performance, employing a two-staged optimization process and utilizing SSE as the objective function without weights. The model exhibits a partial correspondence to the data, leading to relatively small values of the quality metric R^2. This phenomenon is attributable to the greater amount of data at the onset and conclusion of the epidemic. In this experiment, the optimization algorithms tend to provide those values for the parameters that provide lower value of SSE for the bigger number of data points. Therefore, the model fit does not capture peak values of the epidemics.



(a) (b)

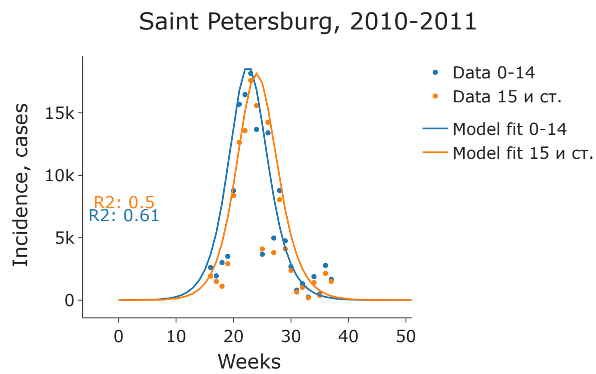
Fig. 3. Model fit over age-specific incidence data of 2010-2011 (left) and 2015-2016 (right) epidemic seasons with weighted SSE



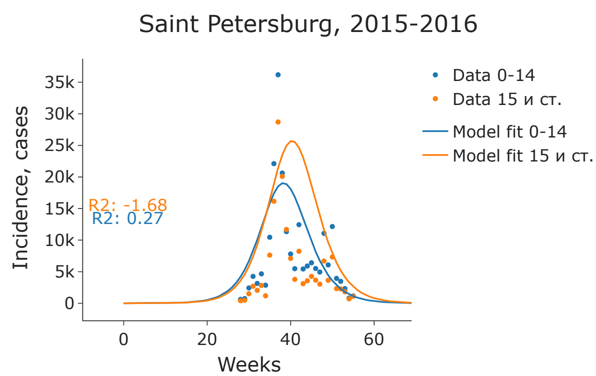
(a) (b)

Fig. 4. Model fit over age-specific incidence data of 2010-2011 (left) and 2015-2016 (right) epidemic seasons without weighted SSE

In an effort to enhance a simulated fit of the model, we introduced weights into the optimization algorithm to apply a higher penalty to the missed peak values. By incorporating these weights, the optimization process becomes more sensitive to accurately capturing the peak of epidemics, which is a critical aspect of modeling infectious outbreaks according to the domain experts. Despite the seemingly reasonable subjective correspondence between the data and the model, there is a decline in the resulting values of R^2. It could be seen in Figure 4.



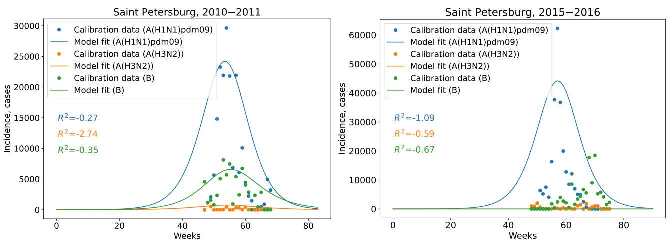
(a)



(b)

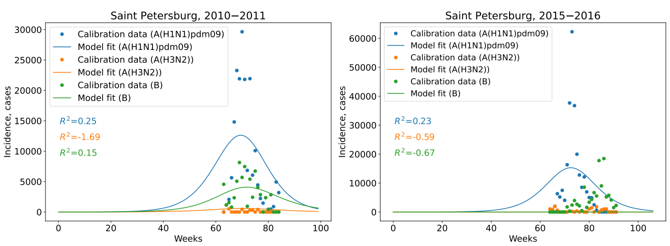
Fig. 5. Manual fitting with GUI of the model over strain-specific incidence data of epidemic seasons 2010-2011 (left) and 2015-2016 (right)

Due to the high non-linearity and multi-modality of the target function, automatic calibration turns out to be computationally intensive process. Throughout this procedure, owing to the aforementioned factors, the model might exhibit a tendency to converge towards a local optimum rather than a global one. This problem is particularly relevant when modeling population dynamics, since the wrong (or late) choice of control measures can lead to monetary costs or human losses. Our GUI designed to facilitate the automatic adjustment of the model with respect to specific parameters while offering experts the capability to fine-tune the model fit through their domain-specific knowledge. Automated adjustment algorithms enable efficient optimization of parameters, ensuring a more accurate representation of epidemic dynamics. Simultaneously, the GUI empowers domain experts to intervene and make informed adjustments based on their specialized insights. By integrating both automatic and manual calibration capabilities, our GUI bridges the gap between computational efficiency and expert-driven decision-making. The results obtained from the GUI demonstrate an improvement in the quality of the model fit for the 2010-2011 season, particularly concerning age-specific incidence data demonstrated in Figure 5 (left). For the season 2015-2016 R^2 increased only for age group up to 14 years.



(a) (b)

Fig. 6. Model fit over strain-specific incidence data of 2010-2011 (left) and 2015-2016 (right) epidemic seasons without weights



(a) (b)

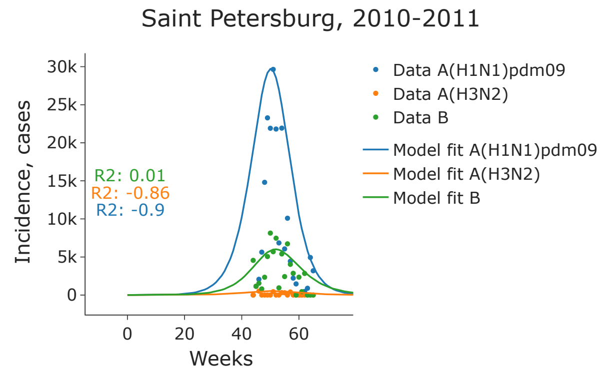
Fig. 7. Model fit over strain-specific incidence data of 2010-2011 (left) and 2015-2016 (right) epidemic seasons without weights

The manual calibration of the model using strain-specific data yielded suboptimal results in terms of quality (Figures 7–8), despite the subjective estimation of the model fit aligning with the epidemic dynamics from the experts' perspective. The strain-specific data presented inherent complexities and variations, making it challenging for automated calibration algorithms to accurately capture the nuances of each strain's transmission dynamics. Consequently, the quality metric, although subjectively perceived as satisfactory by experts, indicated limitations in capturing the underlying complexities of the strain-specific data.

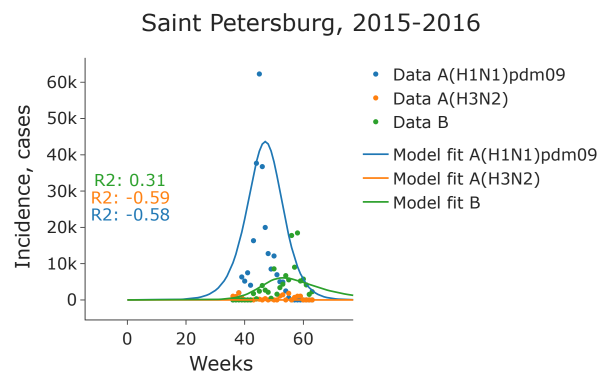
This highlights the significance of human expertise in refining epidemiological models and the need for a balanced approach that combines automated optimization with expert-driven calibration to achieve a more comprehensive and accurate representation of disease transmission dynamics.

V. Discussion

The combined approach of epidemiological model calibration, incorporating both automatic and manual calibration



(a)



(b)

Fig. 8. Manual fitting with GUI of the model over strain-specific incidence data of epidemic seasons 2010-2011 (left) and 2015-2016 (right)

procedures, presents several advantages in achieving a more tailored and accurate representation of disease dynamics, enriched by domain-specific knowledge from experts.

Firstly, the automated calibration process allows for efficient parameter optimization by exploring a wide parameter space, enabling the model to approximate the general trends and patterns of the epidemic. This automated procedure efficiently analyzes a vast range of parameter combinations, facilitating a quick identification of potential parameter sets that yield a reasonable fit to the data.

Secondly, the manual calibration aspect of the approach empowers domain experts to contribute their knowledge and insights. Experts can intervene in the calibration process, fine-tuning the model's parameters to capture specific nuances or peculiarities in disease transmission dynamics. The GUI provides an intuitive platform for researchers and practitioners to interactively calibrate the model using real-world incidence data. We believe that the current version of a framework and its newer and more developed versions might significantly contribute to analysis and modeling of acute respiratory infections thus helping reduce their toll.

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